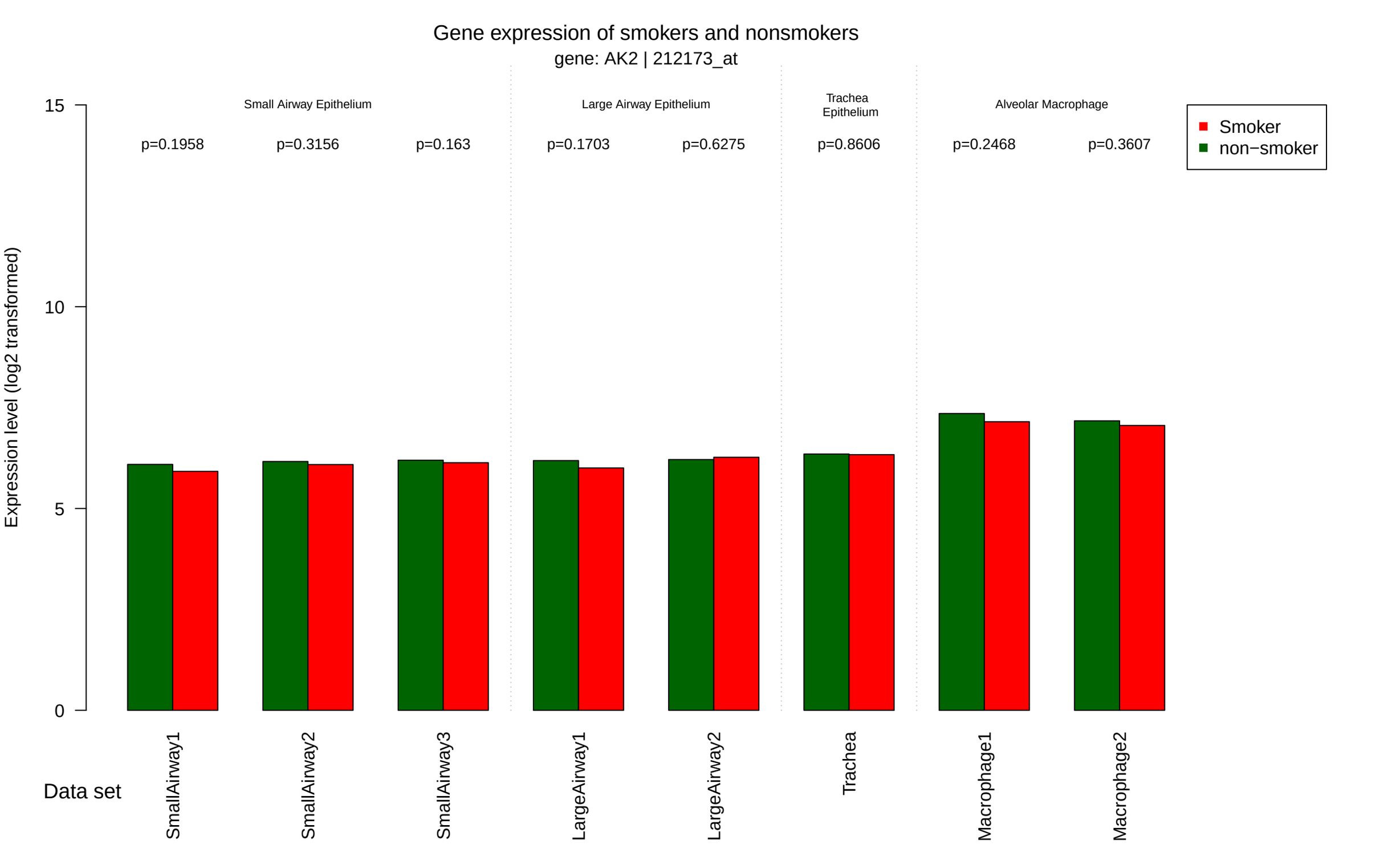
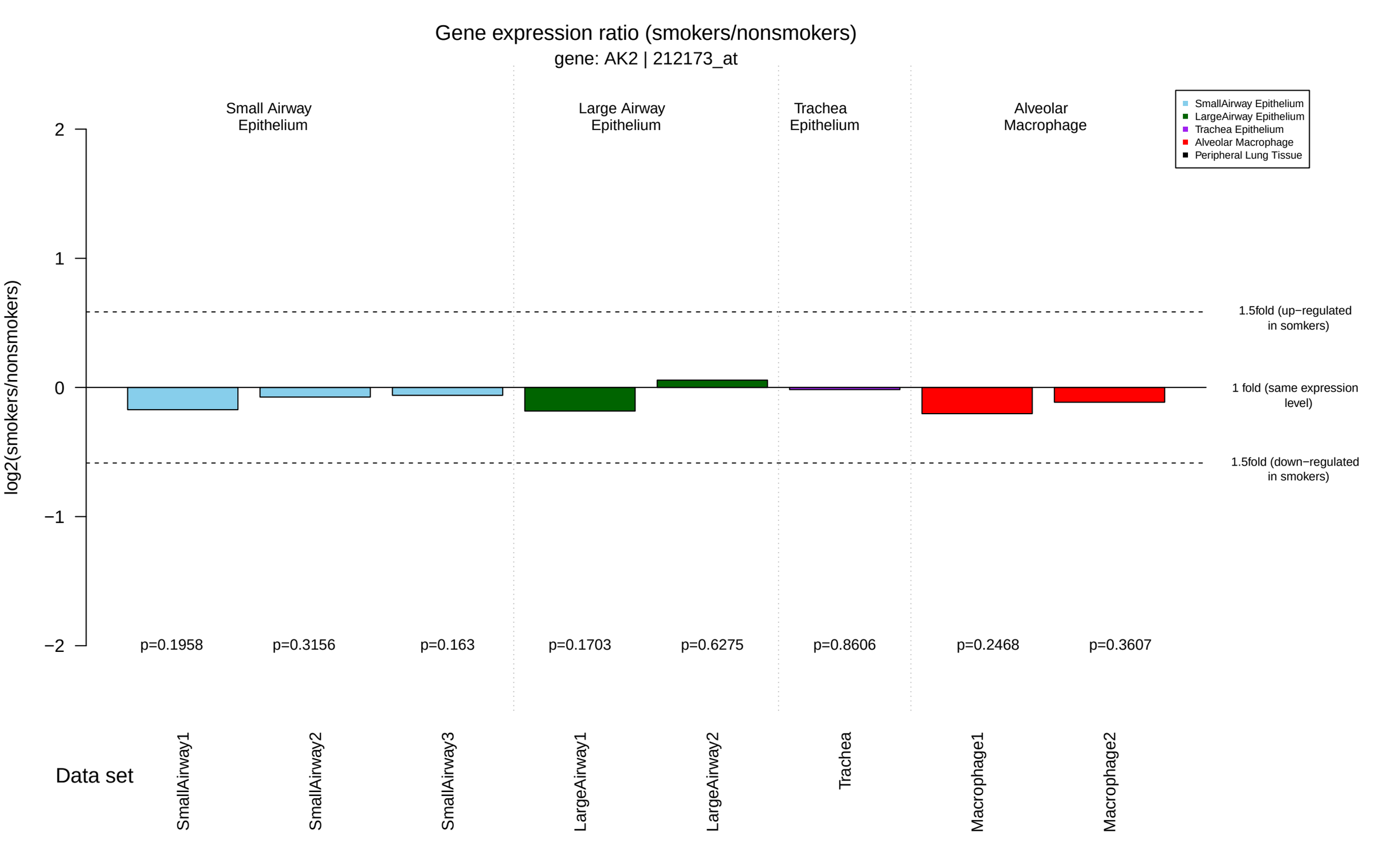
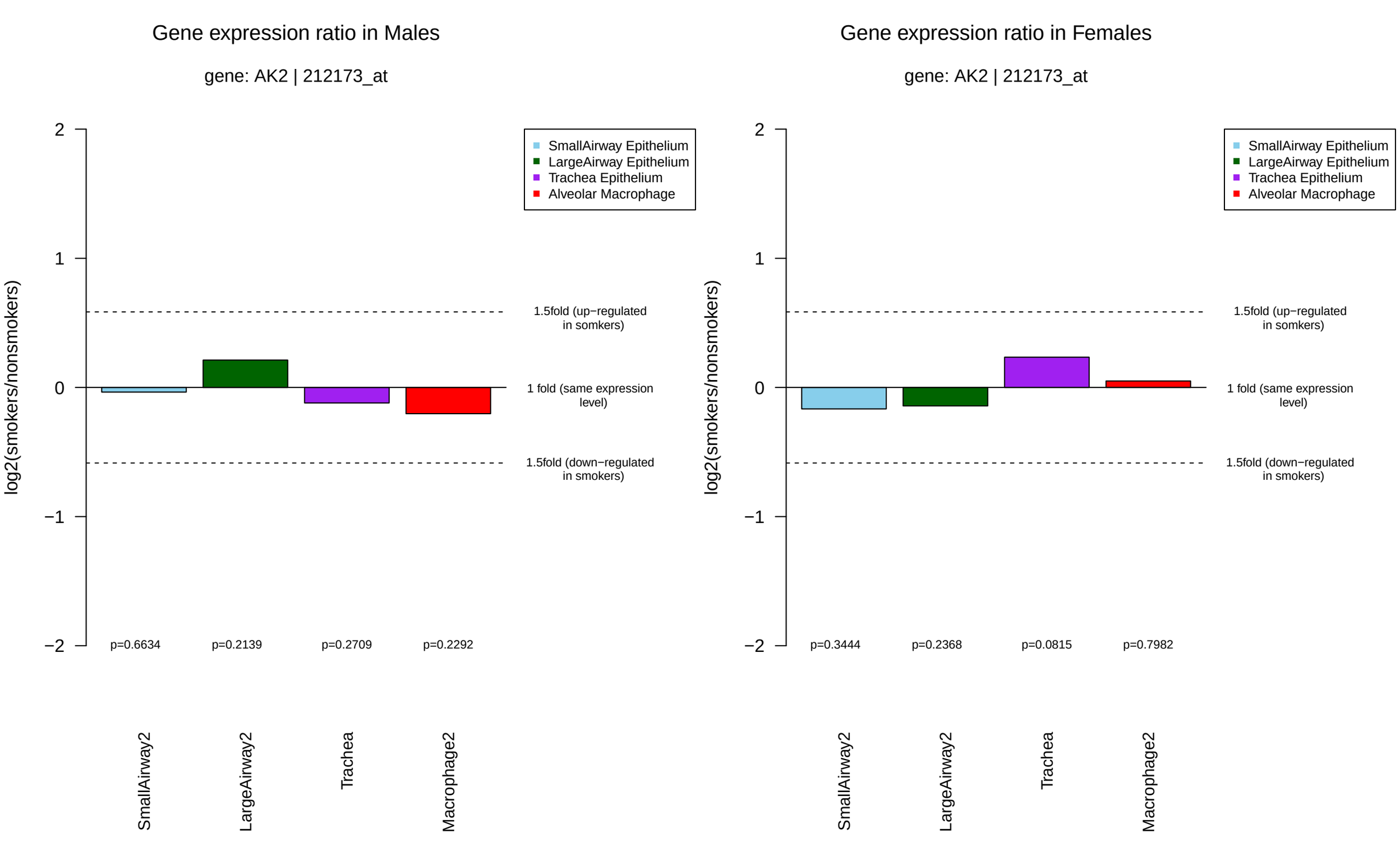
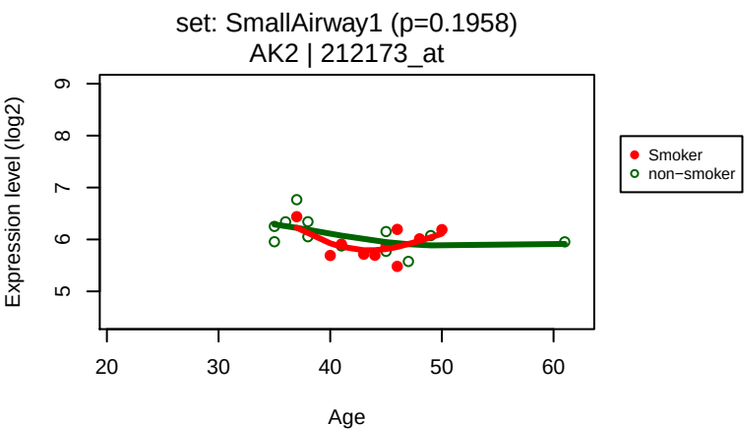
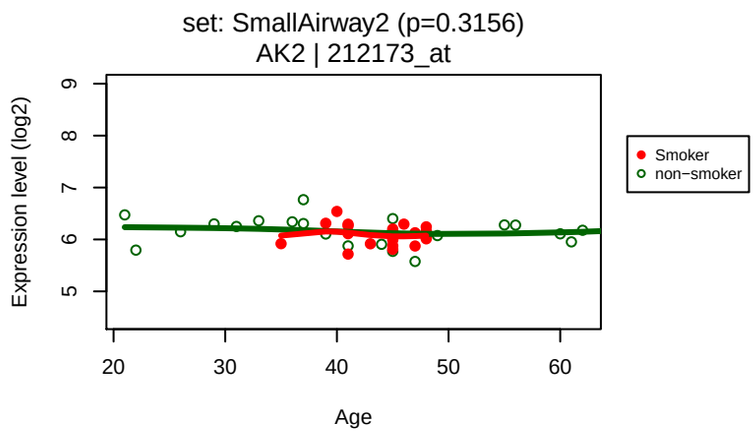
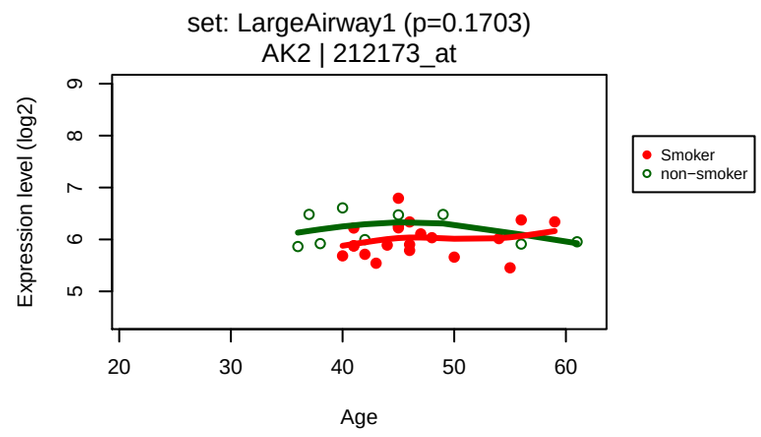

Gene expression of smokers and nonsmokers
gene: AK2 | 212173_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1958
p=0.3156
p=0.163
p=0.1703
p=0.6275
p=0.8606
p=0.2468
p=0.3607
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

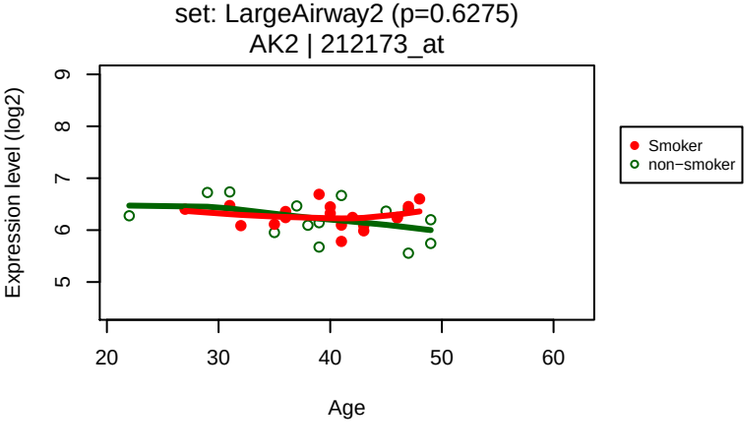

Gene expression ratio (smokers/nonsmokers)
gene: AK2 | 212173_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1958
p=0.3156
p=0.163
p=0.1703
p=0.6275
p=0.8606
p=0.2468
p=0.3607
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

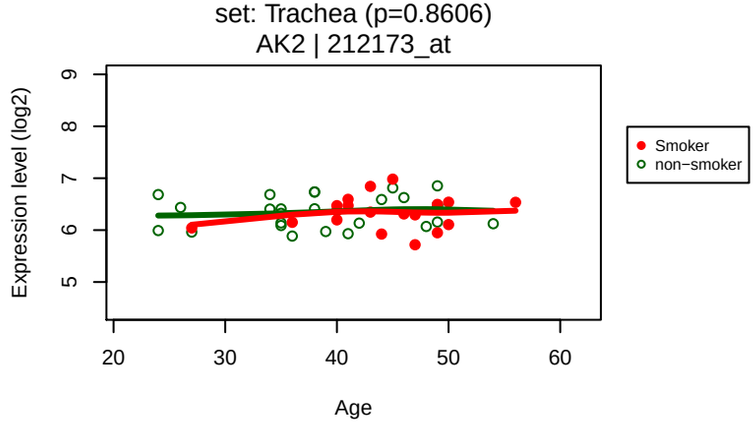

Gene expression ratio in Males
gene: AK2 | 212173_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6634
p=0.2139
p=0.2709
p=0.2292
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK2 | 212173_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3444
p=0.2368
p=0.0815
p=0.7982
SmallAirway2
LargeAirway2
Trachea
Macrophage2

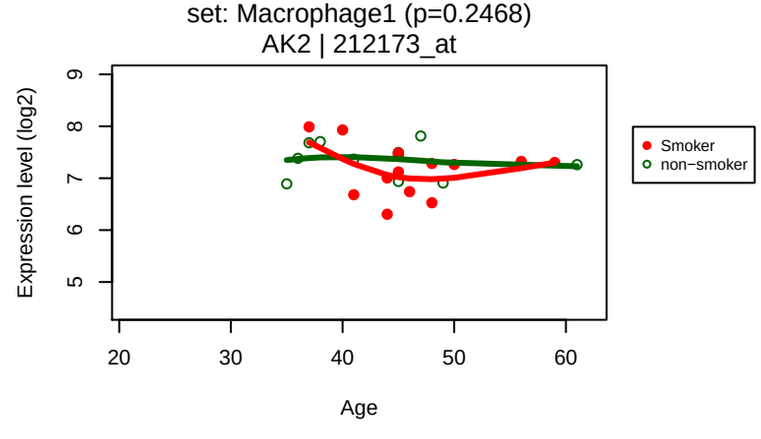

set: SmallAirway1 (p=0.1958)
AK2 | 212173_at
Expression level (log2)
Age
Smoker
non−smoker

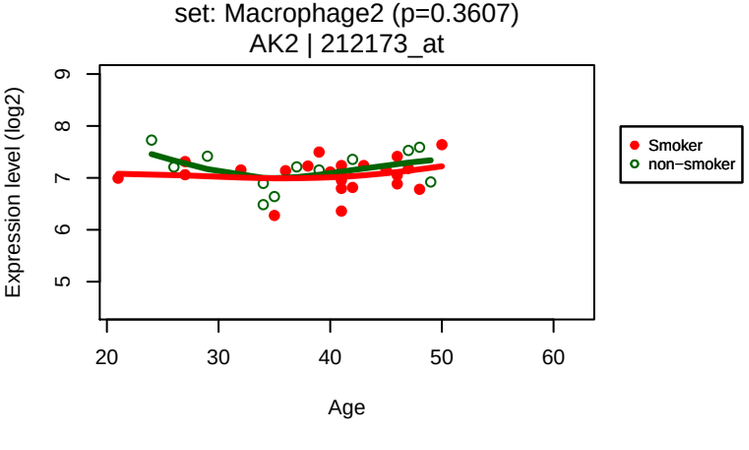

set: SmallAirway2 (p=0.3156)
AK2 | 212173_at
Expression level (log2)
Age
Smoker
non−smoker

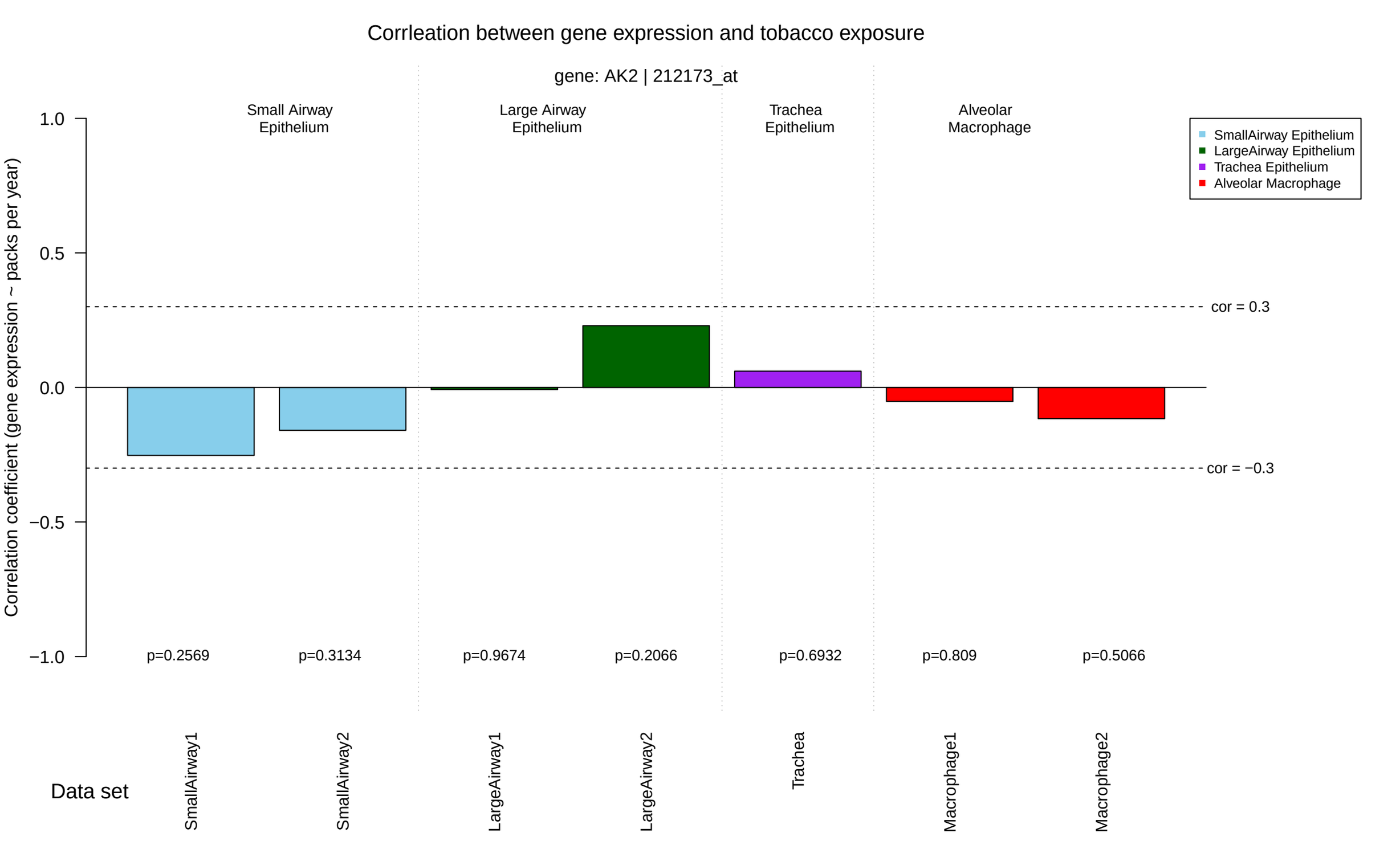

set: LargeAirway1 (p=0.1703)
AK2 | 212173_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6275)
AK2 | 212173_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8606)
AK2 | 212173_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2468)
AK2 | 212173_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3607)
AK2 | 212173_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK2 | 212173_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2569
p=0.3134
p=0.9674
p=0.2066
p=0.6932
p=0.809
p=0.5066
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2